

Complete React Notes for Full Stack Production Projects

These notes cover all essential and advanced concepts of React required for building real-world full stack production applications using MERN stack.

1. React Fundamentals

React is a JavaScript library used for building fast, interactive, and reusable user interfaces using a component-based architecture.

JSX: JavaScript + HTML syntax used inside React components.

Components: Reusable UI blocks. Prefer functional components.

Props: Used to pass data between components.

State: Component's local memory handled using `useState` hook.

2. Hooks

`useState` – Manages local state.

`useEffect` – Handles side effects like API calls.

`useRef` – DOM access and persistent values.

`useMemo` – Optimizes heavy calculations.

`useCallback` – Optimizes function re-creation.

3. React Router DOM

Used for client side routing.

Includes `BrowserRouter`, `Routes`, `Route`, `Link`, `useNavigate`.

Supports protected routes for authentication.

4. Forms Handling

Controlled components.

onChange and onSubmit handling.

Validation using libraries like React Hook Form, Yup, Zod.

5. API Integration

Axios and Fetch API.

Axios interceptors.

Central API service pattern.

6. State Management

Redux Toolkit.

Store, Slice, Reducers.

useSelector and useDispatch hooks.

7. Authentication Flow

JWT based authentication.

Login, Signup.

Protected Routes.

Role based access control.

8. Folder Structure

src/

components/

pages/

api/

hooks/

redux/

routes/

utils/

9. Performance Optimization

React.memo

useMemo

useCallback

Lazy loading

10. Environment Variables

.env usage

process.env

11. Production Build & Deployment

npm run build

Deploy on Netlify, Vercel, AWS.

12. Best Practices

Small components

Reusable UI

Clean architecture

Global error handling

13. MERN Architecture

React → Express → MongoDB → Node

14. Real World Features

Authentication

Pagination

Search

Filters

File upload

Lazy loading

15. Interview Preparation

Virtual DOM

Hooks

Redux

Routing

Performance optimization

This document is designed for students and developers preparing for production level React and full stack projects.